

Chapter 2

Tensors and the Poincaré group

2.1 Raising and lowering indices

We want to map the co-vector construction onto invariants like $p_\mu \eta^{\mu\nu} p_\nu$. To do this, we define a co-vector from a vector by

$$p_\mu = \eta_{\mu\nu} p^\nu \quad (2.1)$$

– that is, by lowering an index with the metric. For the four-momentum, this gives

$$p_\mu = \left(-\frac{U}{c}, \gamma m \vec{v} \right) \quad (2.2)$$

where the minus sign in the first entry arises from $\eta_{\mu\nu}$. Does this object really transform as a co-vector? Note that

$$p^\mu \eta_{\mu\nu} p^\nu = p^\mu (\eta_{\mu\nu} p^\nu) = p^\mu p_\mu \quad (2.3)$$

The left-hand side is invariant, so $\eta_{\mu\nu} p^\nu$ must indeed transform so that the contraction is invariant. To check in general, consider $w_{\mu'} \equiv \eta_{\mu'\nu'} w^{\nu'}$ and write

$$\begin{aligned} w_{\mu'} = \eta_{\mu'\nu'} w^{\nu'} &= \eta_{\mu'\nu'} \Lambda^{\nu'}_{\nu} w^\nu \\ &= \eta_{\mu'\nu'} \Lambda^{\nu'}_{\nu} \eta^{\nu\alpha} w_\alpha \\ &= \Lambda^\alpha_{\mu'} w_\alpha \quad \text{as needed} \end{aligned} \quad (2.4)$$

where the last step follows from

$$\hat{\Lambda}^T \hat{\eta} \hat{\Lambda} = \hat{\eta} \Rightarrow \hat{\Lambda}^T \hat{\eta} = \hat{\eta} \hat{\Lambda}^{-1} \Rightarrow \hat{\eta} \hat{\Lambda}^T \hat{\eta} = \hat{\Lambda}^{-1} \Rightarrow \hat{\eta} \hat{\Lambda} \hat{\eta} = \left(\hat{\Lambda}^{-1} \right)^T \quad (2.5)$$

2.2 Poincaré transformations

The most general transformations preserving the spacetime interval are the *Poincaré transformations*

$$x^{\mu'} = \Lambda^{\mu'}_{\mu} x^{\mu} + b^{\mu} \quad (2.6)$$

consisting of Lorentz transformations – rotations in spacetime – plus translations b^{μ} in spacetime. The Lorentz transformations are given by

$$\hat{\Lambda}^T \hat{\eta} \hat{\Lambda} = \hat{\eta} \quad (2.7)$$

But our transformations so far further require

$$\det \hat{\Lambda} = +1 \quad \text{and} \quad |\Lambda^0_0| \geq 1 \quad (2.8)$$

Full Lorentz symmetry relaxes these conditions

$$\text{Full Lorentz} \rightarrow \text{Lorentz} + \text{parity} + \text{time reversal} \quad (2.9)$$

where parity flips the spatial coordinates, $x^i \rightarrow -x^i$, and time reversal flips $x^0 \rightarrow -x^0$

$$P^{\mu'}_{\mu} = \begin{cases} -1 & \text{if } \mu = \mu' = i \\ +1 & \text{if } \mu = \mu' = 0 \\ 0 & \text{otherwise} \end{cases}, \quad T^{\mu'}_{\mu} = \begin{cases} -1 & \text{if } \mu = \mu' = 0 \\ +1 & \text{if } \mu = \mu' = i \\ 0 & \text{otherwise} \end{cases} \quad (2.10)$$

Under parity and time reversal, the four-velocity transforms as

$$u^{\mu'} = P^{\mu'}_{\mu} u^{\mu} \quad \text{with} \quad d\tau' = d\tau \quad ; \quad u^{\mu'} = -T^{\mu'}_{\mu} u^{\mu} \quad \text{with} \quad d\tau' = -d\tau \quad (2.11)$$

2.3 Tensors

More generally, a *tensor* of rank (p, q) is an object $T^{\mu_1 \dots \mu_p}_{\nu_1 \dots \nu_q}$ that transforms as

$$T^{\mu'_1 \dots \mu'_p}_{\nu'_1 \dots \nu'_q} = \Lambda^{\mu'_1}_{\mu_1} \Lambda^{\mu'_2}_{\mu_2} \dots \Lambda^{\nu_1}_{\nu'_1} \Lambda^{\nu_2}_{\nu'_2} \dots T^{\mu_1 \dots \mu_p}_{\nu_1 \dots \nu_q} \quad (2.12)$$

with one factor of $\hat{\Lambda}$ for each upper index and one factor of $\hat{\Lambda}^{-1}$ for each lower index. In this language: 4-vectors are $(1, 0)$ tensors; co-vectors are $(0, 1)$ tensors; and scalars (like τ) are $(0, 0)$ tensors, that is invariants. The metric provides examples of tensors that do not change under Lorentz transformations

$$\eta^{\mu\nu} \quad \text{of rank } (2, 0) \quad , \quad \eta_{\mu\nu} \quad \text{of rank } (0, 2) \quad (2.13)$$